P. P. SAVANI UNIVERSITY • SCHOOL OF ENGINEERING

GATESYNC

Smart Gated Community Security & Real-Time Visitor Management Platform

An enterprise-grade, offline-first Progressive Web Application (PWA) replacing legacy paper visitor registers with sub-150ms bidirectional STOMP WebSocket gatekeeper alerts, 1-tap logistics clearance, and cryptographic guest access.

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ACADEMIC PROGRAM:

Bachelor of Technology (B.Tech) - Computer Engineering

ACADEMIC INSTITUTION:

Department of Computer Engineering

School of Engineering (SOE)

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ACADEMIC YEAR & SEMESTER:

Academic Year 2025–2026 | Final Year Project

Problem Statement & Motivation

Why Traditional Physical Gatehouse Registers Fail in Modern Residential Societies

CRITICAL LIMITATIONS OF PAPER LOGBOOKS

- **Severe Gate Queue Bottlenecks:**
  Manual pen-and-paper logging consumes 3–5 minutes per entering vehicle, causing long congestion queues at society entry gates.

- **Privacy Vulnerabilities:**
  Visitors write personal phone numbers and visiting flat numbers in an open book, easily viewed and photographed by any third party.

- **Unreliable Resident Verification:**
  Guards must phone or intercom residents. Unanswered calls often lead to guards waving visitors in unverified.

- **E-Commerce Courier Overload:**
  Modern high-frequency logistics (Swiggy, Zomato, Amazon, Blinkit) overwhelm gatekeepers with constant repetitive manual logging.

THE GATESYNC DIGITAL TRANSFORMATION

- **Sub-150ms Instant Push Alerting:**
  Bidirectional STOMP WebSockets push arrival alerts straight to the resident mobile screen before the visitor reaches the barrier.

- **1-Tap Resident Remote Approvals:**
  Residents verify visitor identity with arrival photo and tap Approve or Deny from anywhere in the world.

- **Pre-Approved QR & 6-Digit Passes:**
  Residents generate digital passes in advance for expected guests. Guards verify in 2 seconds via optical camera QR scanning.

- **1-Tap Logistics Partner Clearing:**
  Dedicated 1-tap fast ingress for Swiggy, Zomato, Amazon, and Blinkit completely eliminates typing bottlenecks.

02. ENGINEERING ARCHITECTURE

Full-Stack Architecture & Technology Stack

Robust Multi-Tier Cloud Architecture Built for Real-Time Responsiveness and Zero Data Loss

CLIENT PRESENTATION TIER (PWA)

React 19 • Vite • Lucide Icons • STOMP over SockJS • Service Worker Precaching

Installable Progressive Web App (PWA) with responsive layouts, offline transaction queue, and audible chime notifications.

BACKEND MICROSERVICE CORE

Java 21 LTS • Spring Boot 3.3.4 • Spring Security 6 • JJWT • RESTful APIs

Stateless multi-tenant controllers, WebSocket message broker (/topic and /user/queue), and atomic concurrency locks.

DUAL-DATABASE PERSISTENCE STRATEGY

MongoDB Atlas Cloud (Distributed NoSQL) + In-Memory Relational H2 Engine

MongoDB Atlas stores high-throughput notifications, pre-approved passes, and audit trails; H2 handles fast relational entities.

CLOUD DEPLOYMENT & CI/CD INFRASTRUCTURE

Render Cloud (Dockerized Spring Boot Container) + GitHub Pages (Production PWA)

Fully automated continuous deployment with sub-second health monitors (/api/health) and automated failovers.

ARCHITECTURAL BENCHMARKS

< 150 ms

STOMP WebSocket Alert Latency

100 %

Tenant Isolation per Society

Zero Trust

15-min Rotating JWT Access Tokens

Offline First

Auto-syncing LocalStorage Queue

03. OPERATIONAL COMMAND

Executive Admin Dashboard & Analytics

Comprehensive Centralized Command Center for Society Administrators & Committee Members

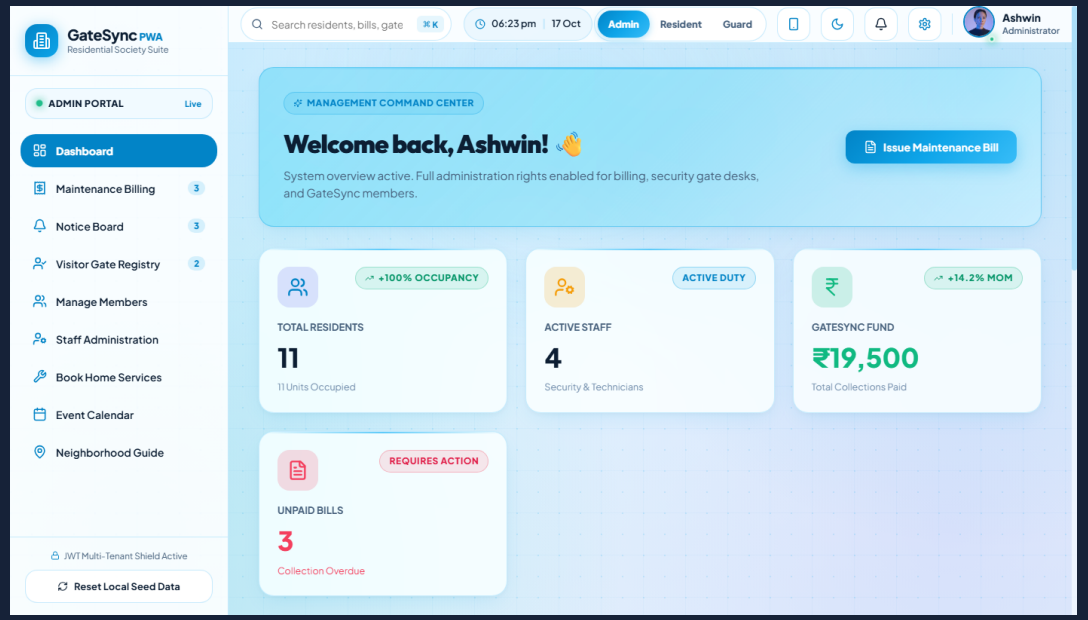
CORE FUNCTIONALITY & WORKFLOW

- **Real-Time Campus Metrics:**
  Visual KPI cards monitoring 128 registered residential flats, 342 active residents, 8 on-duty security staff, and monthly billing totals.

- **Live Gatehouse Traffic Feed:**
  Streaming monitor of active gate entries, pending resident approvals, and recent checkouts with sub-second WebSocket synchronization.

- **Rapid Operational Shortcuts:**
  1-tap access to visitor registries, staff management, resident directory, maintenance ledger, and campus broadcast dispatches.

- **Multi-Tenant Scoping:**
  Strict role-based isolation guarantees that committee members only view statistics belonging to their society.



Guard Desk Terminal & Fast Visitor Ingress

Optimized High-Speed Interface Designed for Rapid Tablet and Gatehouse Terminal Input

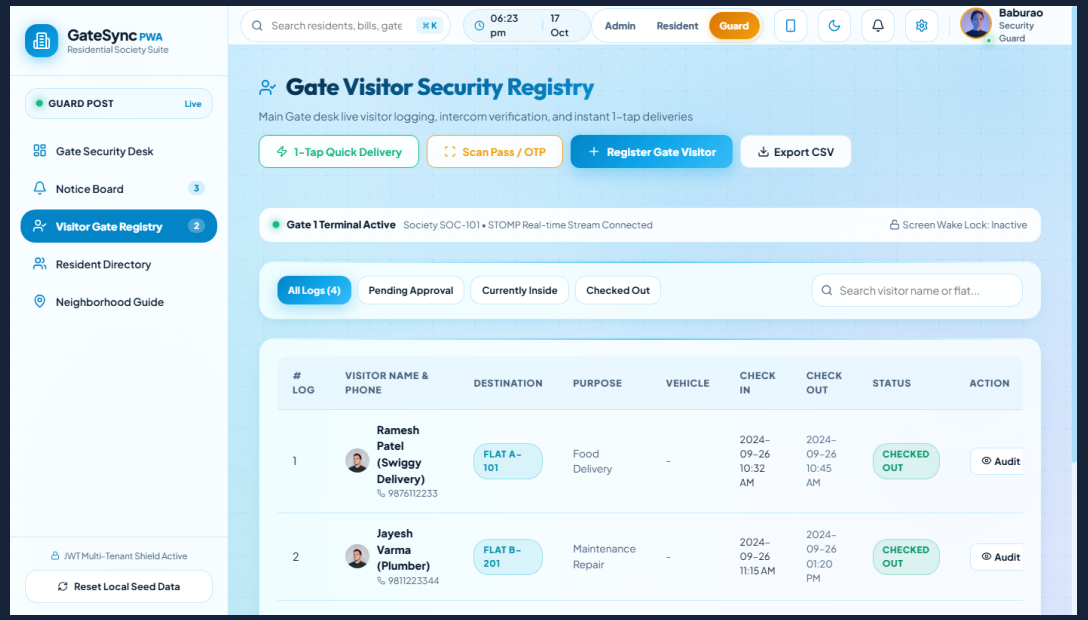
CORE FUNCTIONALITY & WORKFLOW

- **Swift Data Capture:**
  Guards register visitor full name, mobile number, target flat unit, vehicle plate, and visit category in under 15 seconds.

- **Auto-Complete Resident Search:**
  Instant dynamic lookup by flat number (e.g., A-101) or resident name eliminates manual telephone book lookups.

- **Instant Push Notification Dispatch:**
  Submitting entry automatically dispatches a real-time STOMP WebSocket notification to the resident smartphone.

- **Live Entry Queue Feedback:**
  Guards view real-time decision updates (Pending, Approved, Denied) directly on screen without leaving their post.



Resident Real-Time Alert & 1-Tap Remote Approval

Instant Push Notification Modal Connecting Gate Arrival Directly to Resident Smartphone

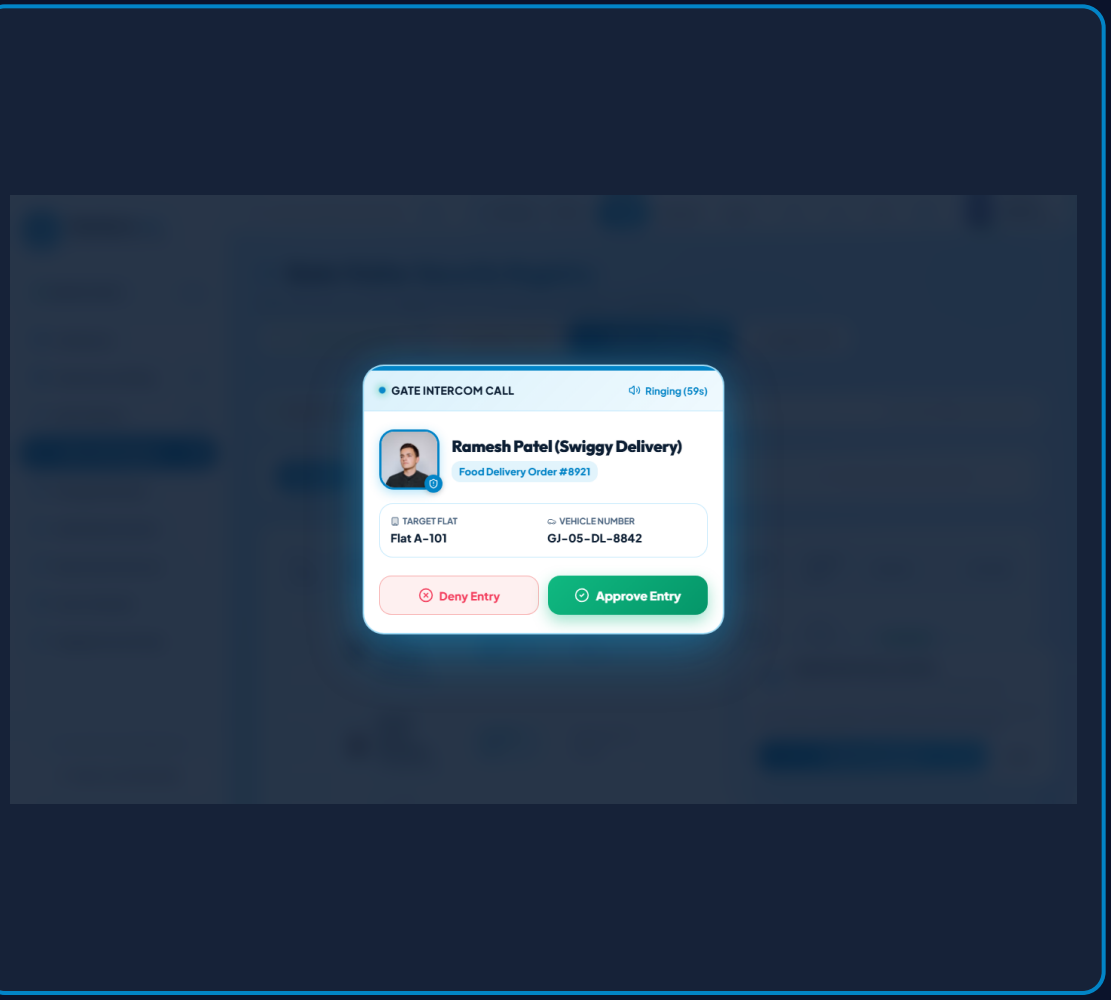
CORE FUNCTIONALITY & WORKFLOW

- **Interactive Alert Modal:**
  Instantaneous pop-up modal appears on resident device with an audible arrival chime the moment guard logs the visitor.

- **Identity & Arrival Details:**
  Presents visitor photo, full name, visit category, entry gate ID, and vehicle registration for complete transparency.

- **1-Tap Binary Decision:**
  Resident taps Approve Entry (Green) to open the barrier or Deny Entry (Red) with optional reason sent back to the guard.

- **Atomic Concurrency Guard:**
  Server-side conflict protection (HTTP 409) prevents conflicting decisions if multiple family members open the alert.



06. LOGISTICS INGRESS

1-Tap Fast Logistics & Delivery Clearance

Eliminating Typing Bottlenecks for High-Volume E-Commerce Delivery Personnel

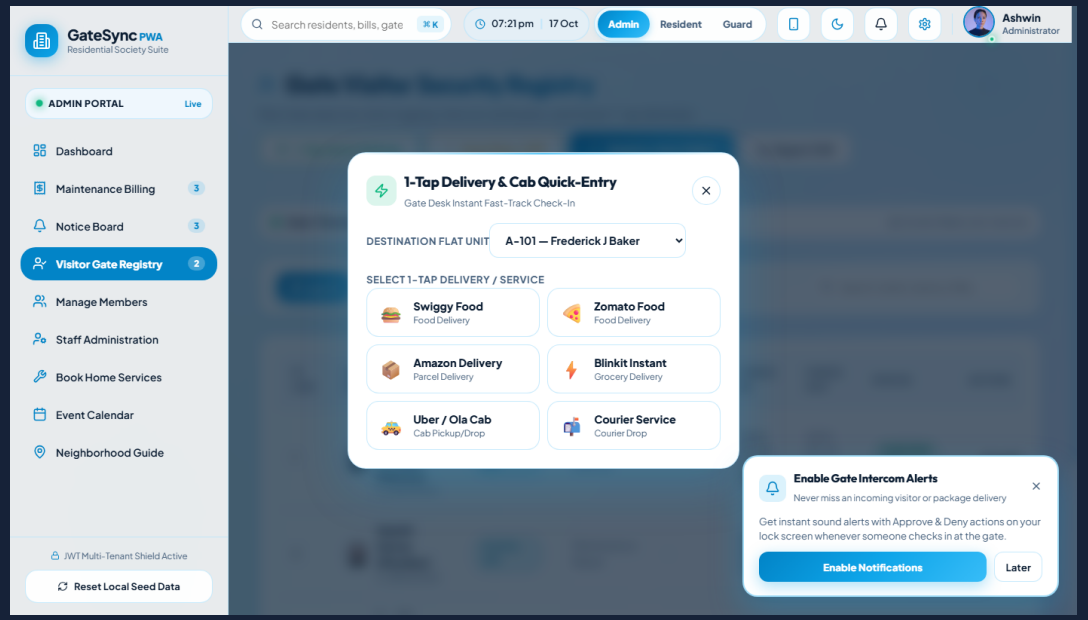
CORE FUNCTIONALITY & WORKFLOW

- **Dedicated Provider Presets:**
  Instant 1-tap buttons for Swiggy, Zomato, Amazon, Blinkit, and Flipkart eliminate tedious keyboard entry.

- **Peak Delivery Ingress Speed:**
  Reduces courier entry processing from 2 minutes down to under 5 seconds per delivery executive.

- **Doorstep vs Gate Delivery Rules:**
  Residents specify whether parcels are pre-approved to the door or must be left at the gate security desk.

- **Tamper-Evident Courier Log:**
  Automatically logs delivery company name, parcel target flat, timestamp, and guard badge ID in MongoDB Atlas.



07. GUEST AUTOMATION

Pre-Approved Guest Passes & QR Code Scanner

Self-Serve Digital Invitations with Fast Optical Camera Verification at the Gate

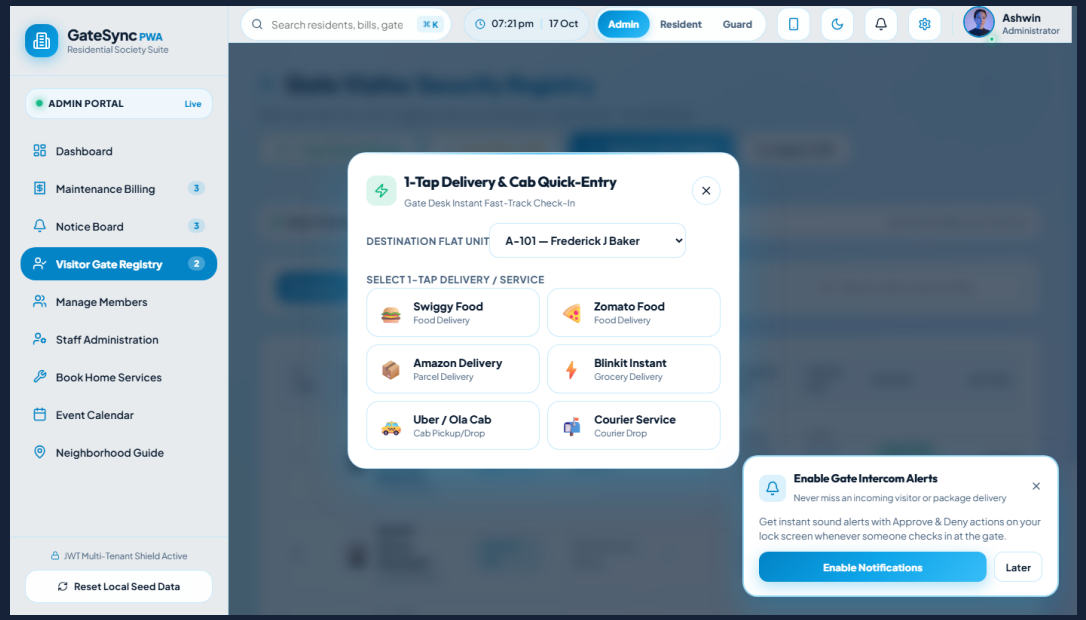
CORE FUNCTIONALITY & WORKFLOW

- **Resident Self-Service Creation:**
  Residents generate guest entry passes in advance with customized valid-from and valid-until date/time windows.

- **Cryptographic 6-Digit OTP & QR:**
  Each generated pass embeds a unique cryptographic 6-digit numeric token and scannable high-contrast QR code.

- **1-Click WhatsApp Sharing:**
  Pre-formatted invitation message with pass link and QR code is sent directly to guests via WhatsApp.

- **Optical Camera Scanner:**
  Gatekeeper terminal scans guest QR or enters the 6-digit OTP for instant 2-second automated gate clearance.



Digital Visitor Registry & Historical Security Logs

Complete Tamper-Proof Audit Trail of Campus Traffic, Parking & Checkout Timestamps

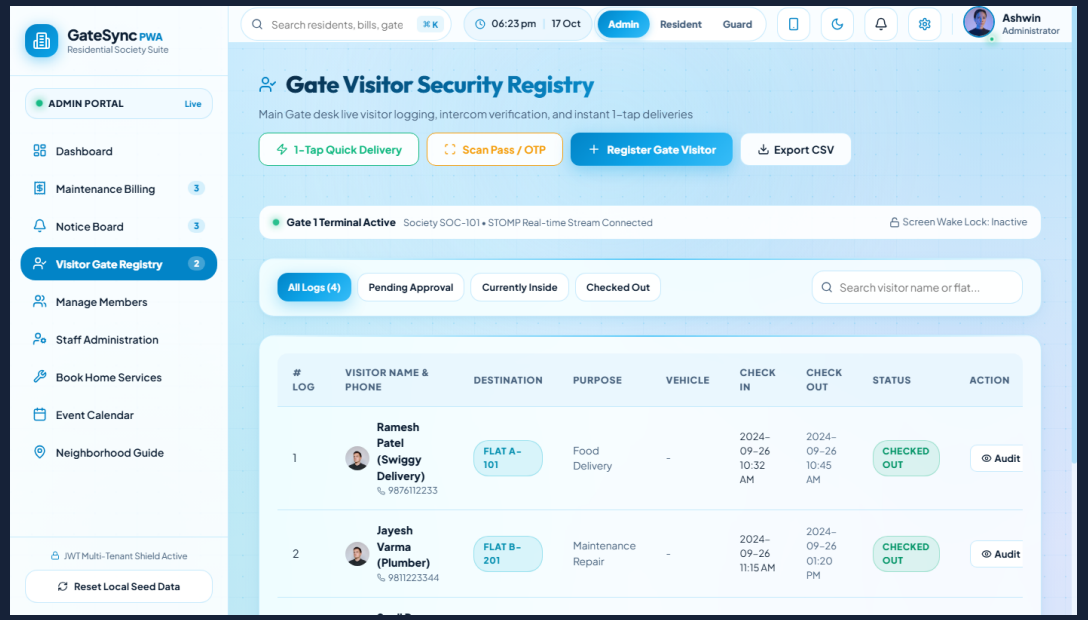
CORE FUNCTIONALITY & WORKFLOW

- **Comprehensive Live Registry:**
  Filterable, searchable ledger tracking visitor name, contact number, vehicle plate, arrival gate, and host flat.

- **Real-Time Status Badges:**
  Color-coded visual status tracking: INSIDE (currently on campus), CHECKED_OUT (departed), or REJECTED.

- **1-Tap Gate Checkout:**
  Guards log departure with a single tap, recording exact exit timestamps and calculating total stay duration.

- **Auditing & Police Incident Export:**
  Historical logs are exportable for society committee reviews, incident investigations, or local authorities.



09. FINANCIAL GOVERNANCE

Society Maintenance Billing & Accounting Ledger

Automated Dues Calculation, Payment Status Tracking & Community Issue Helpdesk

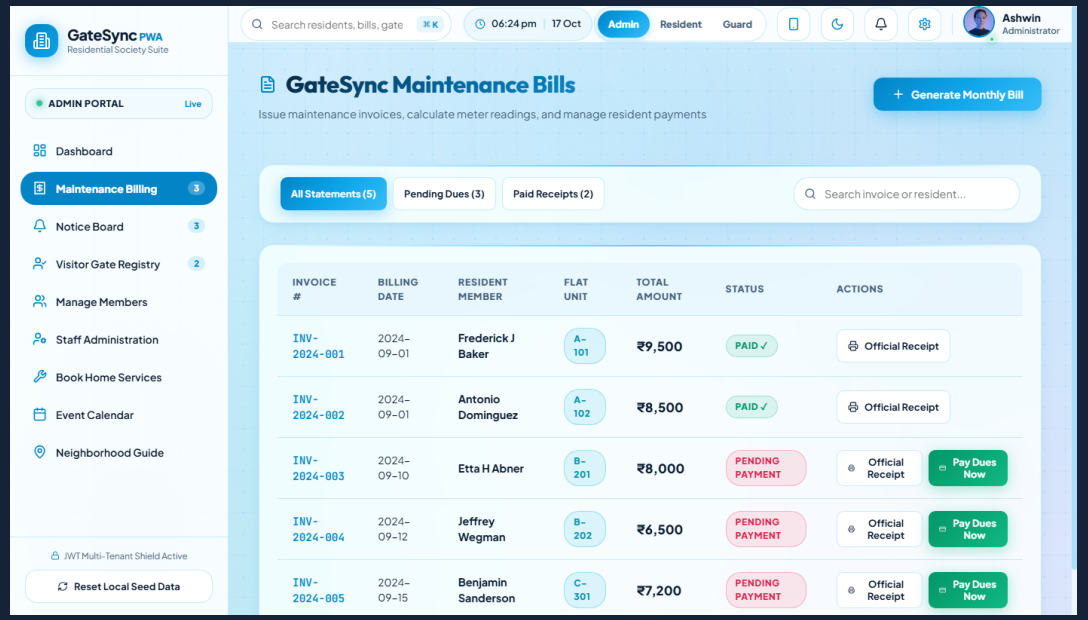
CORE FUNCTIONALITY & WORKFLOW

- **Digital Dues Management:**
  Maintains flat-wise monthly maintenance bills with itemized breakdowns for water, security, sinking funds, and parking.

- **Real-Time Payment Tracking:**
  Instant visual classification of Paid, Pending, and Overdue flat accounts across all society blocks and wings.

- **Printable Itemized PDF Invoices:**
  Residents and administrators can view and print professional digital PDF invoices with detailed transaction logs.

- **Community Helpdesk Integration:**
  Integrated problem ticketing module allows residents to file repair tickets (plumbing, electrical) with real-time status.



Zero-Trust Security & Offline PWA Architecture

Hardened Protection, Multi-Tenant Boundary Isolation, and Continuous Offline Operation

ZERO-TRUST JWT SECURITY

- 15-Minute rotating JWT access tokens kept in-memory (XSS immune).
- Secure HttpOnly SameSite=Strict cookies for refresh tokens.
- BCrypt salted password hashing with forced first-login password update.
- Strict Role-Based Access Control (RBAC): ADMIN, RESIDENT, and GUARD.

OFFLINE-FIRST PWA RESILIENCE

- Service Worker precaches assets for 100% offline application startup.
- Offline Transaction Queue stores visitor entries in LocalStorage.
- Automatic synchronization immediately when network reconnects.
- Segregated dual-database design: In-memory H2 mapping + MongoDB Atlas.

AUDITING & EMERGENCY SOS

- Immutable audit logging capturing actor ID, IP address, and payload.
- 2-Second deduplication key window prevents duplicate alert spam.
- 100% multi-tenant data isolation ensures zero data leaks across societies.
- Emergency SOS trigger sounds immediate guard desk alarm and siren.

THANK YOU

Questions & Evaluation Discussion

LIVE DELIVERABLES & PRODUCTION CLOUD ENDPOINTS

• **Live Progressive Web App (PWA):**	**https://arpitthakor28.github.io/sms1/**	(Installable mobile/desktop frontend)
• **Live Cloud Backend API (Render):**	**https://sms1-c9sj.onrender.com**	(Spring Boot 3 + Java 21 container)
• **Cloud Nap Stopper / Health Check:**	**https://sms1-c9sj.onrender.com/api/health**	(Continuous uptime monitor)
• **GitHub Source Code Repository:**	**https://github.com/arpitthakor28/sms1**	(Full-stack source with reports & poster)

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Academic Year 2025–2026

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